

	KENYATTA UNIVERSITY QUALITY MANAGEMENT SYSTEM	Ref.:	<i>KU/ACAD/SOP/8.5-3</i>
		Ver.:	<i>1.0</i>
Title:	SMA 113: Analytical Geometry	Date:	08/09/2026
	SEMESTER I 2026/2027		

3 Credit Hours

Pre-requisite

None I

Purpose of the Course

To equip the learner with basic geometrical skills and numeracy required in solving real life problems in their fields of study.

Expected Learning Outcomes

By the end of the course, the learner should be able to:

- Sketch graphs of given trigonometric functions specifying the amplitude, phase change and period.
- Derive different trigonometric identities and apply them in solving equations.
- Solve equations involving trigonometric functions and identities.
- Find the equations of conic sections and features related to them.
- Convert given curves from Cartesian to polar or parametric form and vice versa.

Course Content

The straight line: Equation, parallel and perpendicular lines. Distance from a point to a line. The circle: General equation, equations of tangents. Polar coordinates and their relationship with Cartesian coordinates, polar graphs and equations. Conic Sections (parabola, ellipse and hyperbola): Equations in standard, polar and parametric forms, Chords, tangents and normals. Trigonometry: Trigonometric functions, their graphs and inverse for degree and radian measure, Trigonometric identities and equations.

Mode of Delivery

Class lectures, tutorials, discussions, problem solving, exercises.

Instructional Materials

Whiteboard/chalkboard, overhead projector, smart board.

Course Assessment

Continuous assessment tests: 40%

Final examination: 60%

Core Reading Materials

1. Thomas, G.B and Finney, R. L (2004) *Calculus and Analytic Geometry*, 9th edition, Addison – Wesley
2. Zill, D. G. (1988). *Calculus with Analytic Geometry*, Pws Pub Co; 2 Sub Edition.
3. Backhouse, J.K. and Houldsworth, S.P.T (2000). *Pure Mathematics I & II*. London: Longman Group.

Recommended Reading Materials for the Course

1. Skowkowski, E. (2010). *Algebra and Trigonometry with Analytic Geometry*, 2nd edition by Brooks/Cole Publishing Co.

TEACHING SCHEDULE

Week	Topic	Sub-topic
1	The Straight Line	▪ General form of a line; midpoint of a line and distance between two points ▪ Angle formed by intersection of lines
2	The Straight Line	▪ Parallel and Perpendicular lines ▪ Directed and undirected distances from a point to a line
3	Circle	▪ Standard and general equation of a circle; centered origin ▪ Circle centered at any point ▪ Equation of circle given three points
4	Circle	▪ Intersection of circles; circles that touch internally, externally and orthogonally ▪ Equation of tangents and normals to circles (points on the circle) ▪ Equations of tangents from an external point to a circle
5	Parabola	▪ Parabola with vertex at the origin ▪ Parabola with shifted vertex
6	Parabola	▪ Equations of tangents and normals to a parabola ▪ Applications of parabola to real life situations
7	CAT I	
8	Ellipse	▪ Ellipse with center at the origin ▪ Ellipse with center at any point ▪ Equations of tangents and normals to an ellipse

9	Hyperbola	▪ Hyperbola with center at the origin ▪ Hyperbola with center at any point ▪ Equations of tangents and normals to a hyperbola
10	Coordinate Transformations and Trigonometry	▪ Radian measure of angles and relations to degree measure ▪ Transformations of coordinates and equations from Cartesian to Polar forms and vice versa ▪ Trigonometric functions, their graphs and inverse trigonometric functions
11	Trigonometry	▪ Special angles and calculations without use of calculators or tables ▪ Trigonometric identities equations ▪ Solutions of trigonometric equations
12	CAT II	
13 & 14	EXAMS	